Duration of Stock Market Crashes

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Abstract

We study how long it takes the U.S. stock market to recover from a crash, defined as a 10% drop from a recent peak after recovering from the prior crash. Between 1926 and 2024, we identify 36 such events. The average and median recovery durations were 387 and 127 days; the fastest was 30 days, the slowest 4,544. On average, the index fell 28% further before bottoming, taking 13 to 1,039 days. Recovery times vary too much to align with independently distributed returns. This paper studies the factors that influence the duration of these crashes, such as economic downturns, bubbles and crash speed. Using these variables, we construct market timing strategies that produce higher returns and Sharpe Ratios than a buy-and-hold strategy.

Keywords: Stock Market Crashes, Bubbles, Duration, Risk

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1 Introduction

In this paper, we investigate the episodes of stock market crashes in the United States since 1926. We define a crash as a period of time during which the stock market (measured by the CRSP Index) falls by 10% over its recent peak value. 35 episodes of crashes were identified during this period. When looking at the time to recovery for each one of these crashes, we see large heterogeneity. In this paper, we investigate the determinants of this heterogeneity. More specifically, we look to explain why some episodes take too long to recover, as others take shorter than expected. We provide economic and statistical explanations for the causes of these episodes.

We test this by comparing with simulated data using the empirical distribution of returns. Comparing real crashes with similar-sized simulated events, we see that some crashes take significantly longer to recover than what is expected. Most specifically, the crashes that follow the Great Depression of the 1930s, the Dot-Com bubble, and the 2008 financial crisis have a smaller recovery rate. On the other hand, a series of other crashes identified by our methodology are faster to return to their pre-crash levels. We then formulate a simple model for the crash duration which allows us to formulate testable hypotheses for the factors that are linked with longer and shorter crashes. This model is estimated both in structural (via Maximum Likelihood) and reduced form (via Ordinary Least Squares). We find that the presence of economic downturns, such as bubble bursts and downward business cycles, induce longer-lived crashes.

The literature provides several complementary perspectives on the mechanisms that drive stock market crashes and their subsequent recoveries. Huang and Wang (2009) develop an equilibrium model in which crashes arise endogenously from temporary liquidity shortages rather than deteriorating fundamentals. When market participation is costly, order imbalances emerge and generate sharp, asymmetric price declines that are typically short-lived once liquidity returns. By contrast, speculative bubbles lead to more persistent downturns.

Phillips, Shi, and Yu (2012, 2015a,b) introduce recursive econometric tests (the SADF and GSADF procedures) to detect and date multiple periods of explosive behavior in asset prices, showing that the collapses of such bubbles often correspond to major historical crashes in the S&P 500. Consistent with this, Greenwood, Shleifer, and You (2019) document that while sharp price run-ups do not predict low average returns, they substantially increase the probability of a crash, particularly when associated with high volatility, turnover, and stock issuance—factors that also prolong recoveries. Firm-level evidence further highlights heterogeneous crash responses: Wang, Meric, Liu, and Meric (2009) show that high-beta and volatile firms experience deeper losses and slower rebounds, whereas Muir (2017) find that financial crises feature price declines far larger than can be explained by fundamentals. Taken together, these studies suggest that liquidity-driven corrections tend to be brief, while crashes linked to bubbles or financial distress exhibit longer and more uneven recoveries—an idea central to the empirical analysis in this paper.

The following work extends the existing literature by shifting the focus from the occurrence of crashes to the *duration of recovery*. While prior studies model liquidity-driven price drops Huang and Wang (2009) or detect and date speculative bubbles Phillips et al. (2012, 2015a,b) and link run-ups to crash probabilities Greenwood et al. (2019), we show that recovery times display heavy right tails that cannot be explained by i.i.d. returns. Methodologically, we develop a conditional duration model in which recovery time follows an Inverse Gaussian distribution whose parameters depend on observable market conditions estimated through both reduced form and maximum likelihood. This framework links the nature of a crash to its persistence and distinguishes between liquidity-driven corrections and bubble- or recession-induced downturns. Finally, we connect crash duration to post-crash performance, showing that longer crashes are associated with lower Sharpe ratios, thereby integrating the literatures on market liquidity, bubbles, and crisis performance into a unified empirical setting.

This paper is structured as follows: Section 2 exhibits the methodology used to identify the crashes and runs a quick overview of the characteristics of some special events. It also

presents the simulated study that is used to compare the observed with expected duration of such events. Section 3 describes the theoretical model for the duration of crashes and the estimation methods used. Section 4 shows the main findings of this article and Section 5 analyzes market performance and trading strategies. Finally, Section 6 concludes.

2 Data

2.1 Monthly Data Crashes

We analyze the crashes in the CRSP index using monthly data from 1926–2024. Data is obtained from Kenneth French's website. We define crashes as periods in which the market falls by more than 10% from its last month. We identified 31 such events in the data.

For each of these events, we calculate the duration as the number of months it takes for the market to recover from the 10% loss; the total loss to trough and the time to trough. The results are shown in Table 1 in the Appendix B. The duration distribution is highly skewed, with a mean of 24.16 months and a median of 6 months. The maximum observed duration is 180 months, which corresponds to the Great Depression of 1929 in which the market lost 83% of its value.

Using the empirical distribution of the returns, we simulate the duration distribution by drawing *i.i.d.* returns with replacement and calculating the time it takes to reach a 10% gain. Using this method, we calculate the duration p-value for each one of the observed events. The results are shown in the column *duration_pvalue* in Table 1. The p-value is calculated as the proportion of simulated durations that are greater than the observed duration.

From that column, we observe a significant number of long-lived crashes (5) with p-values below 0.05. To test for the significance of that result, we run a binomial test to assess if the observed number of *large* crashes is significantly larger than expected from the *i.i.d.* simulated distribution. Given the number of observed events $n = 31$ and the number of long-lived crashes $\widehat{k} = 5$ under the probability of success $p = 0.05$ the null hypothesis is given by:

$$\mathcal{H}_0 : \widehat{k} \leq np \qquad \mathcal{H}_1 : k > np \tag{1}$$

Using this test, we find a test statistic of 0.16 with a p-value of 0.01, rejecting the null and pointing towards the presence of more long-lasting crashes than expected under the null.

Following the same procedure, we also test whether the data presents crashes that are significantly shorter than expected. This is done by running the same binomial test, but now for the observations in which we observe a duration p-value greater than 0.95. In this case, we find a test statistic of 0.03, with a p-value of 0.79 not rejecting the null hypothesis.

The final results suggest that the observed duration distribution exhibits heavier right tails than expected under the null hypothesis of *i.i.d.* returns. Therefore, some of the crashes we observe are significantly longer. Once we look to these episodes, we observe that these long-lived crashes are mostly associated with large economic events such as the Great Depression of 1929 and the 2000 Dot-Com bubble. The results are shown in Table 1 in the Appendix B.

While the monthly data provide a broad view of long-term market downturns, they may obscure the precise timing and dynamics of crash events, which can unfold within a single month. To address this limitation, we introduce a complementary analysis using daily data. This finer frequency allows for a more accurate identification of crash onset and recovery, as well as a more detailed examination of the returns distribution. By leveraging the higher-resolution data, we aim to capture both the short-term market fluctuations and the structural characteristics that drive the severity and persistence of crashes.

2.2 Daily Data

We analyze crashes in the CRSP index from 1926–2024. We define a crash as a period in which the market falls by more than 10% from its peak, defined in a rolling window of 252 business days (1 year). A total of 36 crashes were found in the data.

Table 2 presents all 36 crash episodes identified in the CRSP index based on daily data from 1926 to 2024. The most severe and long-lasting event corresponds to the Great Depression,

which began after the September 1929 peak and culminated in July 1932 with a total market loss of 182.6%, taking more than 4,500 days to recover. Other major downturns include the 1973–1974 bear market, which lasted 957 days with a 66.2% loss, and the Dot-Com crash starting in March 2000, which extended for 1,641 days and wiped out nearly 69% of the market's value. More recent events also stand out: the Global Financial Crisis (2007–2009), with a cumulative loss of 79.2% over 1,083 days, and the COVID-19 crash in early 2020, which saw a rapid 42% drop but an equally swift recovery within 112 days. The table highlights that, while some crashes were short and sharp—such as those in 1955, 1979, and 2007—others were prolonged and deep, reflecting the diverse temporal and magnitude profiles of major market downturns in U.S. financial history.

Figure 1 displays the path of the market (CRSP) index over our time sample and highlights each one of the identified crashes in a shaded area.

3 Distribution of Crashes

In this section, we study the distribution of crash duration, comparing the realized events against both theoretical and simulated data. For the simulation, we consider both the empirical distribution of returns, assuming an i.i.d. draw from the full sample. For the theoretical distribution, we consider a simple model where the returns follow a Brownian motion with a drift.

We use the duration pdf, the survivor ratio, and the hazard function to compare these distributions. Finally, we perform goodness-of-fit (GoF) tests to assess the capability of the theoretical model to capture the empirical distribution of crash durations.

3.1 Model

Let X_t denote the market security price at time t. Assume that it follows:

$$X_t = X_0 + \int_0^t \mu ds + \int_0^t \sigma dW_s \tag{2}$$

where W_s denotes a Standard Brownian Motion and μ and σ denote the constant drift and volatility of the price process, respectively.

Assume without loss of generality that a crash is detected at time $t = 0$ with $X_0 = 0$. The crash is defined to last as long as the price process is below a certain threshold α relative to its initial value. Duration T then is defined as:

$$T = \min\{t : X_t \geq \alpha\} \tag{3}$$

Following Lancaster (1990), we know that the distribution of T is given by an Inverse Gaussian distribution with parameters (α, μ, σ) and probability density function given by:

$$f(t) = \frac{\alpha}{\sigma\sqrt{2\pi t^3}} \exp\left\{-\frac{(\alpha - \mu t)^2}{2\sigma^2 t}\right\} \quad \text{for } t \geq 0 \tag{4}$$

The moments of the duration distribution are given by:

$$\mathbb{E}(T) = \frac{\alpha}{\mu} \qquad \mathbb{V}(T) = \frac{\alpha\sigma^2}{\mu^3} \tag{5}$$

3.2 Simulated Data

We start by simulating alternative paths for the market index using the empirical distribution of returns. We assume the returns are i.i.d. and randomly draw them from the full sample with replacement. We simulate 100,000 crashes using the empirical distribution. This is done by simulating a path for the market index until we observe a crash. We then record the time it takes for the index to recover to its initial value[1]. Along with that, we fit the Inverse Gaussian distribution explained in Section 3.1 to the simulated data. We estimate the parameters (μ, σ) using the method of moments and set $\alpha = 0.10$.

Figure 2 shows the estimated density for the simulated crashes against the realized ones. We see that the real data presents heavier tails than both the proposed fits, with a higher density on the right tail. Moreover, both the simulated and theoretical distributions show higher density on low durations not seen on the real data, with the simulated crashes density being slightly heavier like the observed one. This suggests that the crashes shown in the real data tend to be more persistent than the ones proposed by the models.

Aligned with that, Figure 3 shows the hazard function for each of these distributions. The hazard function of a distribution with CDF F and PDF f is defined as $f(x)/(1 - F(x))$ and denotes the probability of leaving the crash state at the next instant given that the crash has sustained until the x. Comparing the hazard function for each of these distributions, we see that the empirical data presents heavier tails than both the simulated and theoret-

[1]The iteration is reset if no crash is detected (or recovered) after 50,000 days

ical distributions, due to the higher likelihood of longer crashes in this dataset. On the left side, the theoretical and simulated data show a higher likelihood of recovery than the empirical data, suggesting that crashes would last shorter in these models. Comparing the behavior of the theoretical and simulated distributions, we see a similar pattern, with the theoretical simulation still showing a heavier right tail. These results show the inefficiency of the two proposed models to capture the distribution of crash durations. The proposed distributions show different likelihood for both longer and shorter crashes when compared with the empirical data.

In the next section, we evaluate the proximity of the distributions by applying goodness-of-fit tests.

3.3 Goodness-of-Fit Tests

To assess the capability of these models to fit the data, we apply the Kolmogorov-Smirnov (KS) test to the densities. Since the moments of the Inverse Gaussian distribution are unknown, we use Gunes, Dietz, Auclair, and Moore (1997) methodology to calculate the critical values. The results are shown in Table 3. The results show that the IG model fails to capture the distribution of duration, with a significance level lower than 1%. Moreover, there is a significant difference between the simulated and theoretical distribution, implying that, even on the assumption of i.i.d. returns, the model would not be able to correctly describe the longevity of crashes.

This creates a challenge for modeling these events. In this paper, we extend the model to include the possibility of factors altering the distribution of returns. This will be explored in Section 4

4 Duration Factors

Following the results of Section 3, we extend the model to allow for external factors affecting duration. This is an easy extension that enables us to generate heterogeneities in the duration distribution while inducing testable hypotheses to be applied to the data.

Consider again a crash detected at time $t = 0$ and let $X_0 = 0$ without loss of generality. We assume that there are external factors, denoted by F that determine the distribution of returns during the crash. By external, we mean these factors are known, or predictable, at the time the crash is detected. Moreover, for the sake of identification, we assume that these factors vary between crashes.

In that case, the return path during the crash will be denoted as a function of F through its parameters in the following way:

$$X_t = X_0 + \int_0^t \mu(F)ds + \int_0^t \sigma(F)dW_s \tag{6}$$

Given F then, we can rewrite the distribution of the duration T as:

$$f(t \mid F) = \frac{\alpha}{\sigma(F)\sqrt{2\pi t^3}} \exp\left\{-\frac{(\alpha - \mu(F)t)^2}{2\sigma(F)^2 t}\right\} \quad \text{for } t \geq 0 \tag{7}$$

with moments given by:

$$\mathbb{E}(T \mid F) = \frac{\alpha}{\mu(F)} \qquad \mathbb{V}(T \mid F) = \frac{\alpha\sigma(F)^2}{\mu(F)^3} \tag{8}$$

4.1 Estimation

Denote a series of observed durations and factors by $(T_i, F_i)_{i=1}^N$. To enable us to estimate the above model, we consider that both μ and σ are linear functions of F:

$$\mu(F) = \beta' F \qquad \sigma(F) = \gamma' F \tag{9}$$

these parameters can then be estimated using a pseudo maximum likelihood – by assuming that the conditioning information variables that prevailed at the start of the crash (i.e., when stock index dropped by 10% for the first time from the immediately preceding peak) affect the mean and the variance of the parameters governing the lognormal distribution of stock index returns – but those parameters remain constant till the stock index recovers from the crash – obviously this is a restrictive assumption – we test whether the dependence on the conditioning variable does not exist and find that we can reject that hypothesis – we do this by testing if the 36 sets of conditional mean and conditional variances are the same – we do this using likelihood ratio test where we constrain all the parameters to be the same across the 36 crashes.

The ML estimator is given by $\left(\hat{\beta}, \hat{\gamma}\right) \coloneqq \arg\max_{(\beta,\gamma)} \ell(\beta, \gamma)$ where ℓ is the log-likelihood function given by:

$$\begin{aligned}
\ell(\beta, \gamma) &= \sum_{i=1}^{N} \log f(T_i \mid F_i) \\
&= \log \alpha - \log\left(\sigma(F_i)\sqrt{2\pi T_i^3}\right) - \frac{(\alpha - \mu(F_i)T_i)^2}{2\sigma(F_i)^2 T_i} \\
\ell(\beta, \gamma) &\propto -\log\left(\gamma' F_i\right) - \frac{\left(\alpha - \left(\beta' F_i\right) T_i\right)^2}{2\left(\gamma' F_i\right)^2 T_i}
\end{aligned} \tag{10}$$

The Appendix C shows the necessary and sufficient conditions for the consistency and asymptotic normality of this maximum likelihood estimator.

Alternatively, we can use the Generalized Method of Moments (GMM) to estimate these parameters knowing the moments of the Inverse Gaussian distribution. The GMM estimator of (β, γ) is given by $\left(\hat{\beta}, \hat{\gamma}\right) = \arg\min_{(\beta,\gamma)} \sum_{i=1}^{N} \left(g_i(\beta, \gamma) - \hat{g}_i(\beta, \gamma)\right)^2$ where

$$g_i(\beta,\gamma) = \begin{pmatrix} \frac{\alpha}{\beta' F_i} \\ \frac{\alpha(\gamma' F_i)^2}{(\beta' F_i)^3} \end{pmatrix} \qquad \hat{g}_i(\beta,\gamma) = \begin{pmatrix} T_i \\ \left(T_i - \frac{\sum_{i=1}^{N} T_i}{N}\right)^2 \end{pmatrix} \tag{11}$$

4.2 Results

In this section we present the results on the estimated models shown in the previous section. We start with the reduced form model, where we regress the observed duration against a set of factors observed before the drawdown. We then present the results for the conditional model estimated using Maximum Likelihood.

For all these models, we include the following set of regressors: the *speed* of the crash, measured as the number of days from peak to the 10% loss defining the drawdown; the level of the *CAPE* index from Shiller (2015) at the time of the peak; the *recession* identifier, measured as a dummy variable on whether we detect an NBER business cycle within the crash period. Although not known at the time of the crash, the *recession* indicator is included in an attempt to control for economic decline crashes and strengthen the findings of the other variables. In order to achieve predictability of this model, one can use recession forecasts such as the FED Recession Indicator – although this is not available for the entire sample of this exercise. Finally, we include the *bubble* identifier. For this, we use the test by Phillips et al. (2015a) (PSY) that allows us to estimate explosive behaviors in the time series. This test is based on a recursive calculation of the Augmented Dickey-Fuller (ADF) test statistic of:

$$\Delta y_t = \alpha + \beta y_{t-1} + \sum_{i=1}^{k} \gamma_i \Delta y_{t-i} + \epsilon_t \tag{12}$$

Denoting by $ADF_{t_1}^{t_2}$ the ADF statistic of from Equation (12) from t_1 to t_2, the Backward

Supremum ADF statistic of PSY is defined as:

$$BSADF_t = \max_{t_1 \leq t_2 - \delta} ADF_{t_1}^{t_2} \tag{13}$$

A bubble is finally identified as starting at T_0 and ending at T_1 if:

$$T_0 = \arg\min_t \{BSADF_t > cv_t\} \qquad T_1 = \arg\min_{t > T_0 + \tau} \{BSADF_t < cv_t\} \tag{14}$$

By applying this method recursively, we can identify multiple bubbles in the time series and time when they start and burst. We detect the bubbles in the monthly *CAPE* index and create a dummy variable that takes the value of 1 if the respective peak date is within a bubble period.[2] For this estimation, we consider a minimum window (δ) of 50 months, a minimum bubble length (τ) of 6 months and critical values at the 95% level.[3]

4.3 Reduced Form

We first analyze a reduced form model, and regress the duration of the observed crashes against a set of factors observed before the drawdown. The covariates used in this analysis are: the *speed* of the crash, measured as the number of days the market takes to lose 10% of its peak value; the level of *cape* from Shiller (2015); the identifier of a *recession*, measured as a dummy variable that takes the value of 1 if the respective peak is within a recession period using the NBER business cycle data; and finally the *bubble* identifier, taking the value of 1 if the respective peak is within a bubble period using the Phillips et al. (2015a) methodology.

Results are shown in Table 4. The coefficients point towards the importance of variables such as the NBER recession and bubbles indicator being positive and significant. More

[2]The estimation of the PSY test is done using the `psybubbles` module in python, available on https://github.com/joseparreiras/psybubbles.

[3]The critical values are calculated using extensive Monte Carlo simulations, available in the package, with 10.000 iterations of window size equal to 10.000 months.

importantly, the interaction of both variables appears as large and significant in the last equation, suggesting that crashes following a bubble period take even longer to recover. For some regressions, the *CAPE* was also significant implying a longer duration the higher the overpricing of the market during the peak. Controlled for that, there does not seem to be a significant effect from the *speed* of the crash.

4.4 Conditional Duration Model

We now estimate a *conditional*, parametric model in which the expected return and volatility during a crash depend on observable state variables measured at the onset of the crash. The goal is empirical: to assess whether market conditions prevailing when the drawdown is first detected help explain subsequent recovery times. We do not take a stand on structural mechanisms; instead, we test for state dependence in reduced-form parameters governing crash dynamics.

Under the maintained assumption that returns during a crash are i.i.d. and log-normally distributed, the implied distribution of recovery times is Inverse Gaussian with parameters (α, μ, σ), where α is the recovery threshold, μ is the mean return, and σ is the return volatility. We allow both μ and σ to vary linearly with the vector of conditioning variables F:

$$\mu(F) = \beta' F \qquad \sigma(F) = \gamma' F. \tag{15}$$

We estimate (β, γ) jointly by maximum likelihood. This specification implies that the conditional mean and volatility are fixed within each crash episode from detection until recovery.[4]

Estimation results are reported in Table 5. The table presents coefficients (and standard errors) for the conditional mean and for the conditional volatility in separate columns, expressed in percentage terms and annualized (mean multiplied by 252; volatility multiplied

[4]Explicitly allowing for time-varying moments would require specifying a transition path for μ_t and σ_t, which is beyond the scope of this paper.

by $\sqrt{252}$).

The covariates F coincide with those used in the reduced-form analysis: (i) *speed* of the crash (days from peak to the 10% drawdown threshold); (ii) the *CAPE* valuation measure of Shiller (2015) at the peak; (iii) an *recession* indicator based on NBER business cycle dating; and (iv) a *bubble* indicator constructed using the Phillips et al. (2015a) procedure.

Table 5 delivers patterns consistent with the reduced-form evidence but with a clearer decomposition into mean-versus-volatility channels. In our estimates, *speed* loads positively on the conditional mean and is not statistically different from zero in the volatility equation, indicating that faster-onset crashes are followed by higher expected returns. The *recession* indicator is statistically significant in both equations, pointing to meaningful shifts in conditional moments during recessionary crashes. The *CAPE* variable enters with negative loadings in both the mean and volatility equations, while the *bubble* indicator primarily affects volatility. The interaction of *bubble* and *recession* further modifies volatility during crash episodes.

To gauge magnitudes, we translate the estimates into effects on expected duration. Figure 4 reports the marginal effect of each covariate on $\mathbb{E}[T \mid F]$, evaluated at the sample mean of the remaining regressors:

$$\frac{\partial \mathbb{E}[T \mid F]}{\partial F_j}(\bar{F}) = \frac{\alpha}{\beta' \bar{F}} \beta_j. \tag{16}$$

The x-axis shows the covariate level and the y-axis reports the implied change in expected duration (in days) for a unit change in that covariate, with 95% confidence bands constructed from the estimated asymptotic variance–covariance matrix.

5 Market Performance

This section analyzes how the stock market performs following crash episodes and how the characteristics identified in Section 4 relate to subsequent return and risk dynamics. The goal here is to interpret how the market's conditional return distribution varies with observable factors at the onset of crashes.

Figure 5 reports the average excess return of the market over the risk-free rate during each crash episode. Across events, returns exhibit substantial dispersion and high volatility, as indicated by the wide confidence intervals. This elevated volatility, however, is not systematically associated with higher mean returns, suggesting that periods of heightened uncertainty do not necessarily compensate investors with higher premia.

Figure 6 summarizes these dynamics in terms of Sharpe Ratios, with bubble size proportional to crash duration. A negative relationship emerges between duration and the Sharpe Ratio: longer crashes tend to be associated with lower risk-adjusted returns. This pattern is intuitive—prolonged drawdowns are typically accompanied by persistent deteriorations in fundamentals and lower expected growth, while short-lived corrections often correspond to transient liquidity or sentiment shocks that quickly revert. Short crashes thus exhibit temporarily elevated Sharpe Ratios, frequently exceeding 2 on an annualized basis.

To quantify the implications of these patterns, we evaluate hypothetical market exposures conditional on crash duration. Specifically, we contrast two benchmark allocations: (i) a "post-crash exposure," which holds the market from the onset of a crash until its recovery; and (ii) a "crash-avoidance exposure", which stays out of the market during identified crashes and re-enters only after recovery. A simple comparison of risk-adjusted performance across these allocations indicates that avoiding crash periods yields superior average Sharpe Ratios, consistent with the notion that prolonged drawdowns destroy rather than create value.

5.1 Factor-Conditioned Return Dynamics

Building on the estimates from Section 4.4, we now examine how observable crash characteristics relate to subsequent market performance. Figure 7 depicts the marginal effects of each conditioning variable on the realized Sharpe Ratio. The results indicate that *speed* is positively associated with higher risk-adjusted returns—faster crashes tend to be followed by stronger recoveries. In contrast, the presence of a *bubble* significantly depresses the Sharpe Ratio, while *recession* periods exhibit a negative though statistically weaker relationship.

We formalize this relation by constructing conditional performance portfolios that vary exposure according to the state variables. For each crash event, we sort the conditioning variable (*speed*, *bubble*, or *recession*) into quartiles based on its historical distribution. If the variable lies in the quartile associated with positive expected performance (e.g., high *speed*), the portfolio takes a long position in the market; if it lies in the adverse quartile (e.g., presence of a *bubble*), the position is either reduced or switched to the risk-free asset. These exposures are not designed as implementable trading strategies but rather as analytical devices to measure conditional differences in market performance.

Table 6 reports descriptive statistics for each conditional exposure, both with and without short-selling constraints. The results indicate that the market performs systematically better when crashes occur outside of bubble or recessionary conditions. The *bubble* and *recession* signals yield significantly positive αs relative to the market and exhibit lower betas, indicating partial hedging properties. Without short selling, the conditional Sharpe Ratios improve from a full-sample market value of 0.364 to 0.458 for the *bubble*-conditioned and 0.469 for the *recession*-conditioned exposures.

Finally, we assess the statistical significance of these conditional performance differences using the test of Cumby and Modest (1987), which extends the Henriksson and Merton

(1981) market-timing framework. The test estimates the regression:

$$R_t^M - R_t^{RF} = \alpha + \beta X_t + \gamma' F_t + \varepsilon_t, \tag{17}$$

where $R_t^M - R_t^{RF}$ is the market excess return, X_t represents the conditional exposure, and F_t includes lagged returns to account for autocorrelation. Statistical significance of β corresponds to the presence of predictable conditional market exposure effects.

The results, reported in Table 7, confirm modest but significant evidence—at the 10% level—that the *bubble*-conditioned exposure explains variation in excess returns. These findings suggest that conditioning on macro-financial states, particularly bubble indicators, is informative about the cross-sectional variation in market risk-adjusted performance following crash episodes.

6 Conclusion

A Images

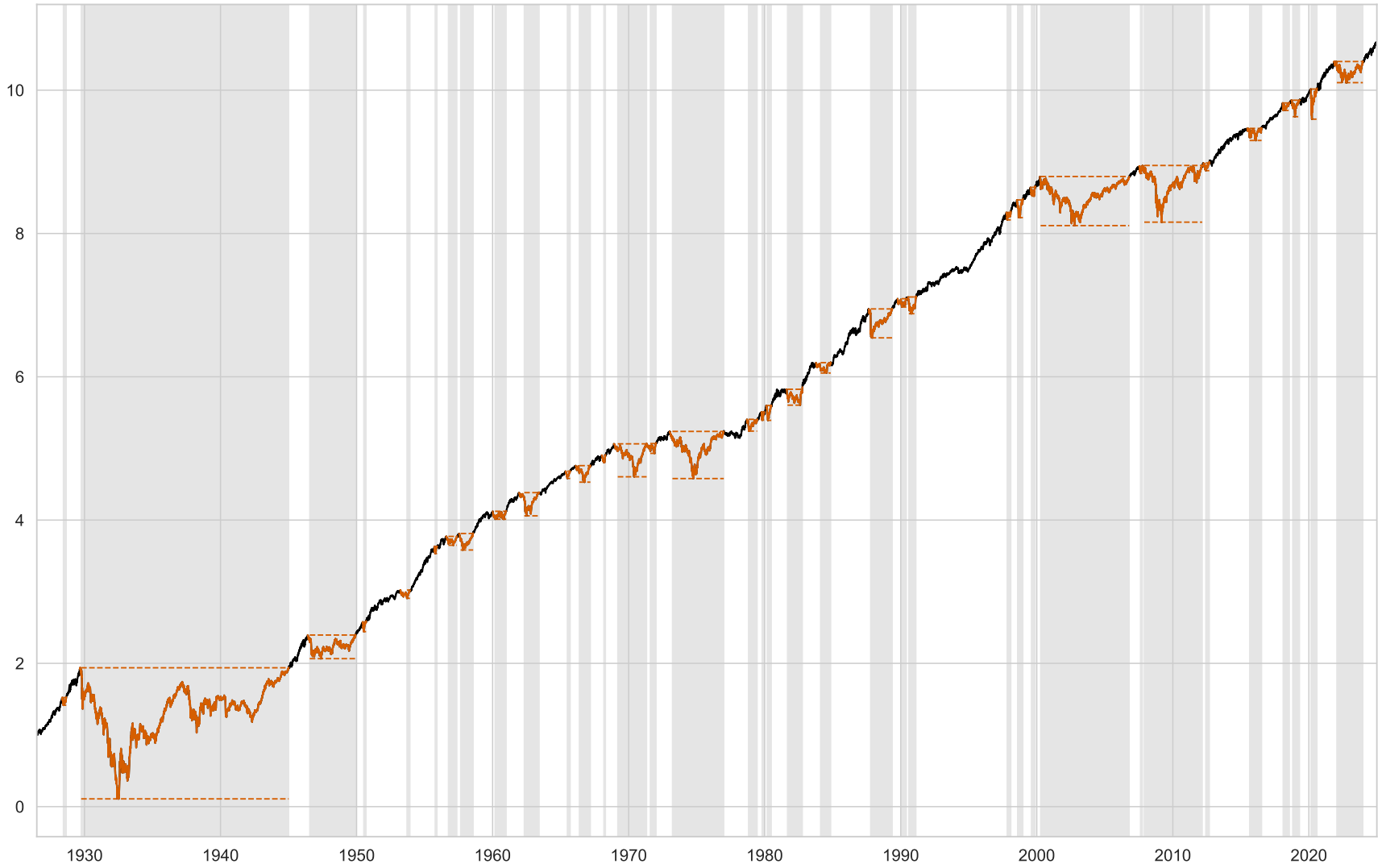


Figure 1: Crashes in the CRSP Index

Notes: Time series of the CRSP Index. Shaded areas represent the crashes, identified as periods in which the index falls from its last 252 business days peak by more than 10%, ending when the index returns to its pre-crash peak level. The dotted red lines show the minimum and maximum values of the index during each crash.

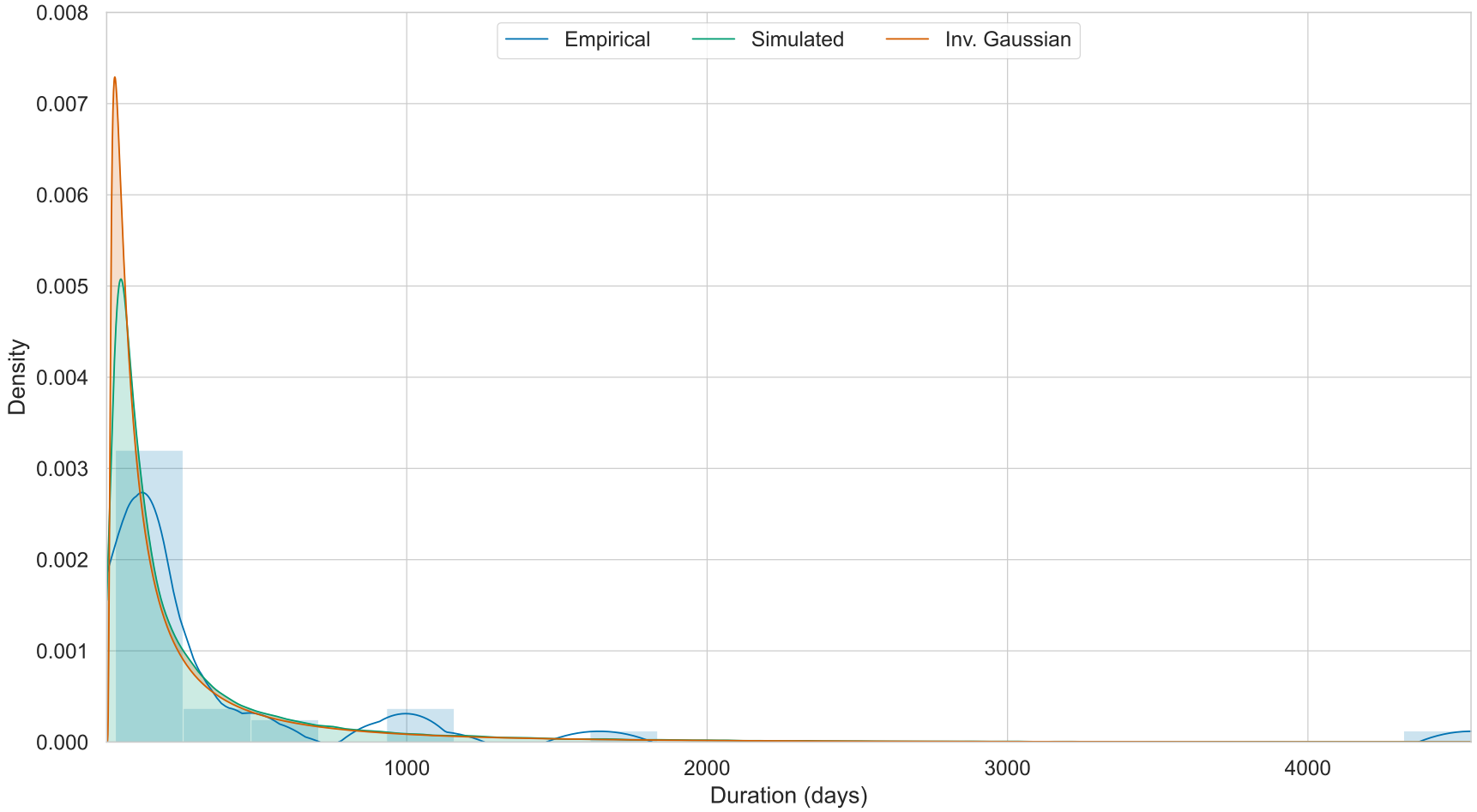


Figure 2: Duration Distribution – PDF (Empirical vs. Simulated vs. Inverse Gaussian)

Notes: Probability density function of duration. The lines represent the empirical distribution, calculated using kernel density estimation using the observed durations in the time series; the simulated data, generated using *i.i.d.* draws of the empirical distribution of returns and calculating the time to reach a 10% gain from the peak (set at 0); and the Inverse Gaussian distribution, estimated using the Method of Moments using the empirical mean and variance of the returns.

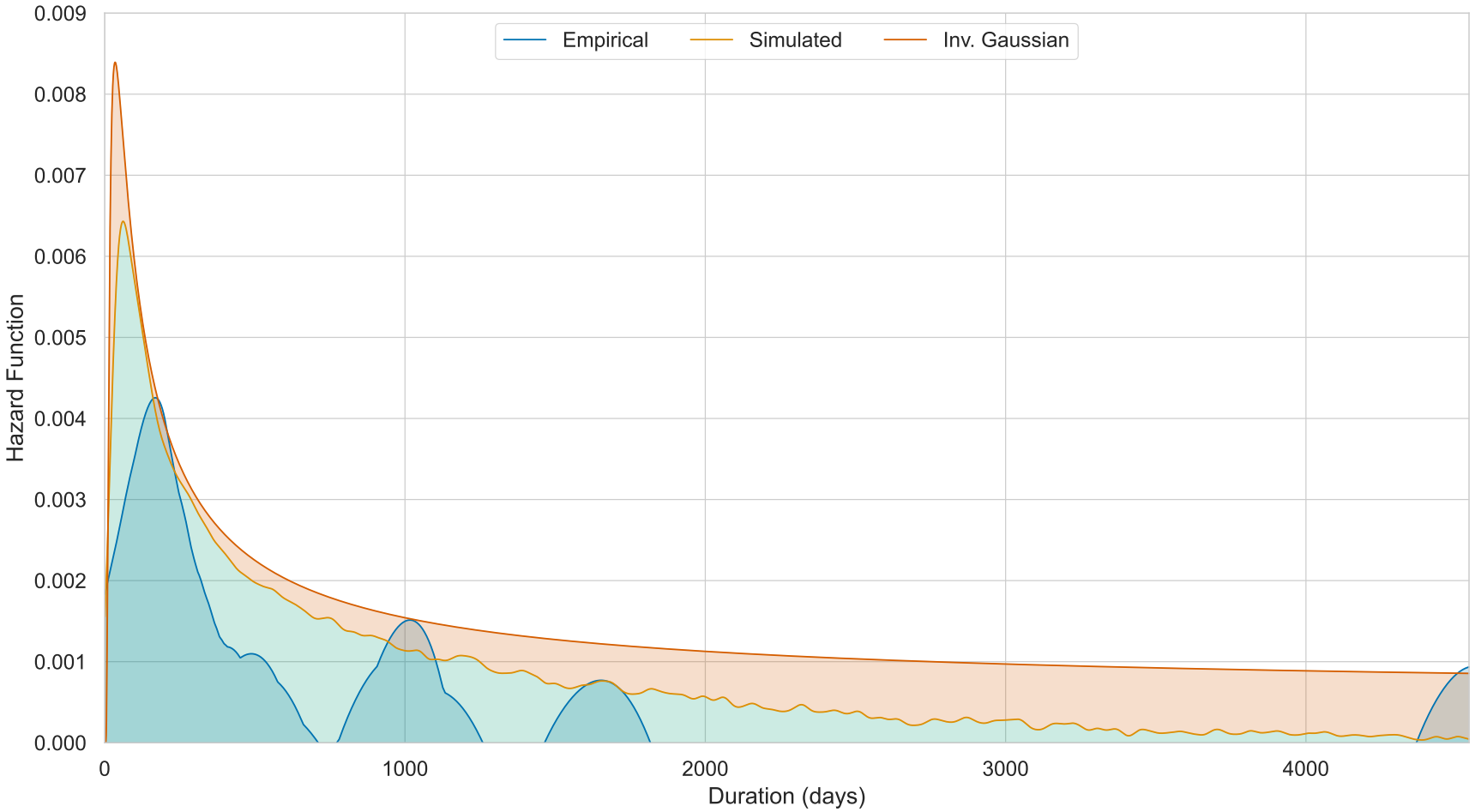


Figure 3: Duration Distribution – Hazard Function (Empirical vs. Simulated vs. Inverse Gaussian)

Notes: Hazard function of duration. Defined as the probability of leaving the crash in the next day, conditioned on being on it at the current time. The lines represent the empirical distribution, calculated using kernel density estimation using the observed durations in the time series; the simulated data, generated using *i.i.d.* draws of the empirical distribution of returns and calculating the time to reach a 10% gain from the peak (set at 0); and the Inverse Gaussian distribution, estimated using the Method of Moments using the empirical mean and variance of the returns. The hazard function is then calculated as $f(x)/(1-F(x))$ for each of these distributions.

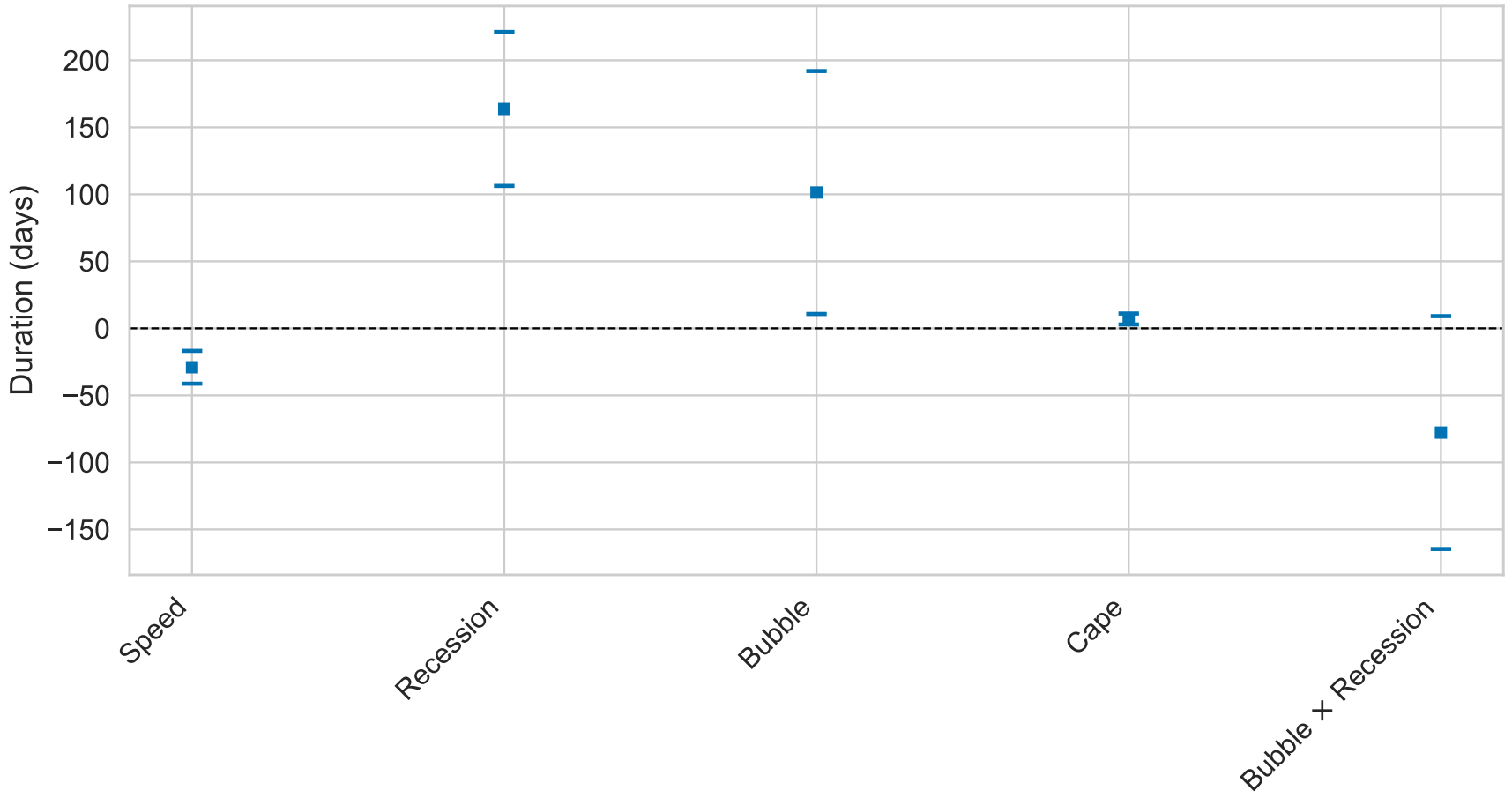


Figure 4: Marginal Effect on the Mean of the Covariates on the Duration (MLE)

Notes: Marginal effect of each of the covariates on the duration of the crashes, estimated by maximum likelihood. The model contains the following covariates: the *speed* of the crash, measured as the number of days the market takes to lose 10% of its peak value; the level of *cape* from Shiller (2015); the identifier of a *recession*, measured as a dummy variable that takes the value of 1 if the respective peak is within a recession period using the NBER business cycle data; and finally the *bubble* identifier, taking the value of 1 if the respective peak is within a bubble period using the Phillips et al. (2015a) methodology; *bubble_and_recession* represents the interaction of both dummies. The marginal effect is calculated as the partial derivative of the expected duration with respect to each of the covariates, conditional on the mean value of all other variables. The x-axis shows the value of the covariate, while the y-axis shows the effect on the duration (in days) for a unit increase in that variable. The error bars represent the 95% confidence interval of the marginal effect calculated by drawing 1000 samples from the asymptotic distribution of the parameters using the estimated variance-covariance matrix.

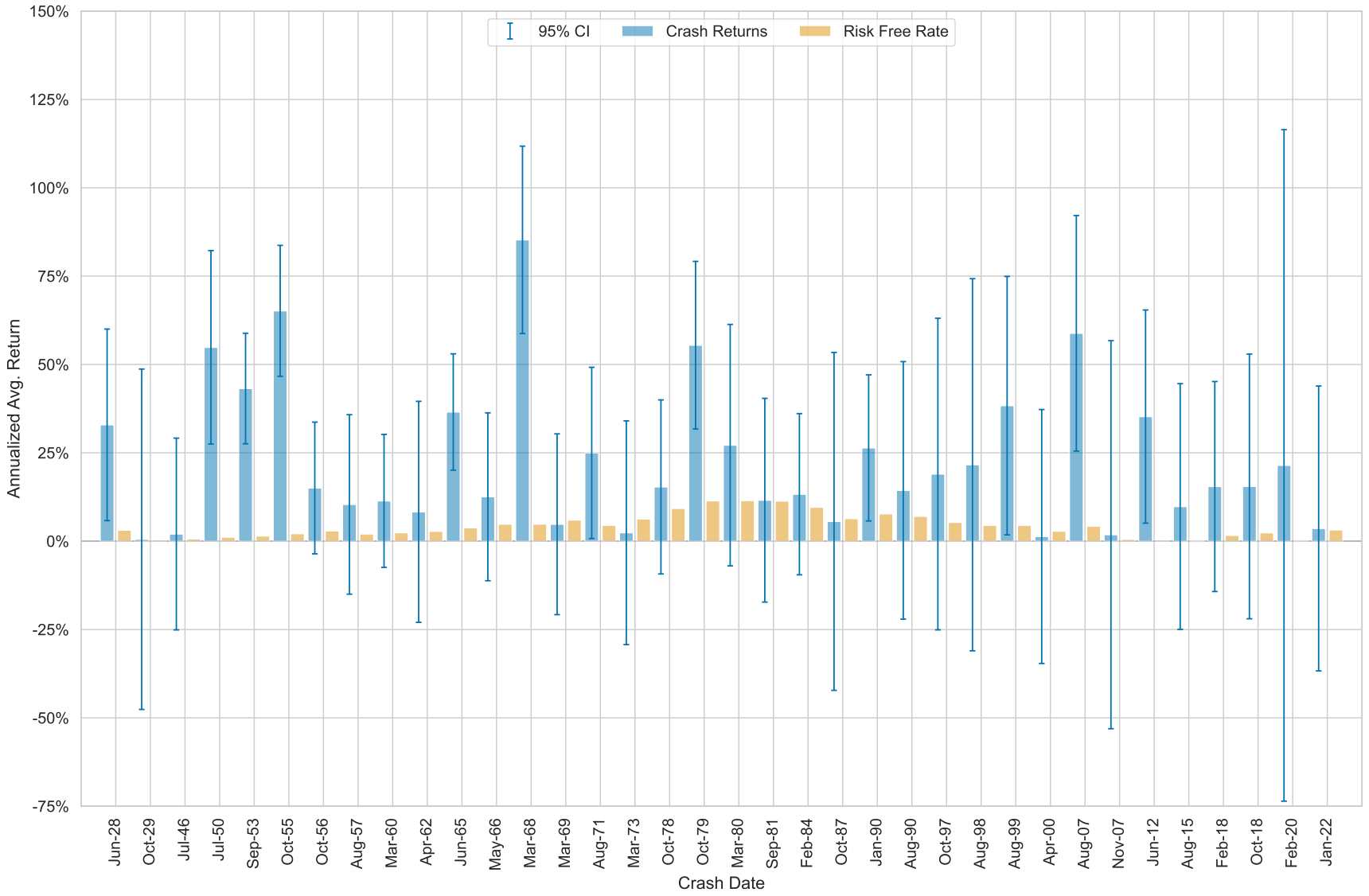


Figure 5: Crash Returns vs. Risk-Free Rate

Notes: Financial performance of the index during each of the crashes. The blue bars represent the average returns during the crash periods (multiplied by 252 to annualize). The blue error bars represent the 95% confidence interval, calculated as 1.96× the standard deviation of returns (multiplied by $\sqrt{252}$ to annualize) in each of the crashes around the mean. Finally, the orange bars represent the (annualized) returns of the risk-free rate during the same periods for comparison. For the x-axis, the labels represent the start date of the crash, given by the date in which the index falls by more than 10% from its last peak.

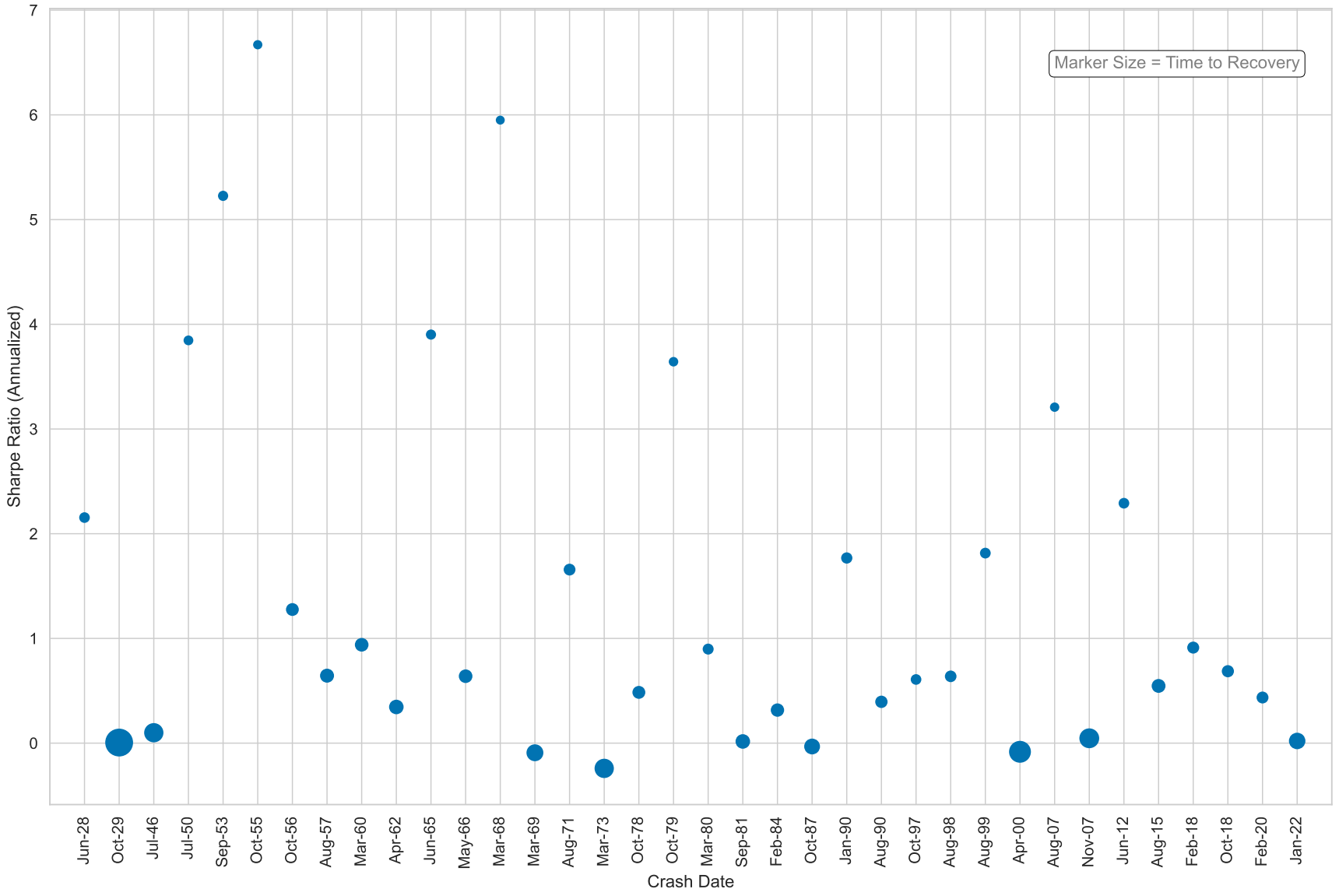


Figure 6: Sharpe Ratio during Crashes

Notes: Sharpe Ratio of the index during each crash event. Calculated as the average return divided by the standard deviation of returns during each crash. The final value is then multiplied by $\sqrt{252}$ to annualize. The circle size is proportional to the duration of the crash, with bigger circles representing longer crashes. For the x-axis, the labels represent the start date of the crash, given by the date in which the index falls by more than 10% from its last peak.

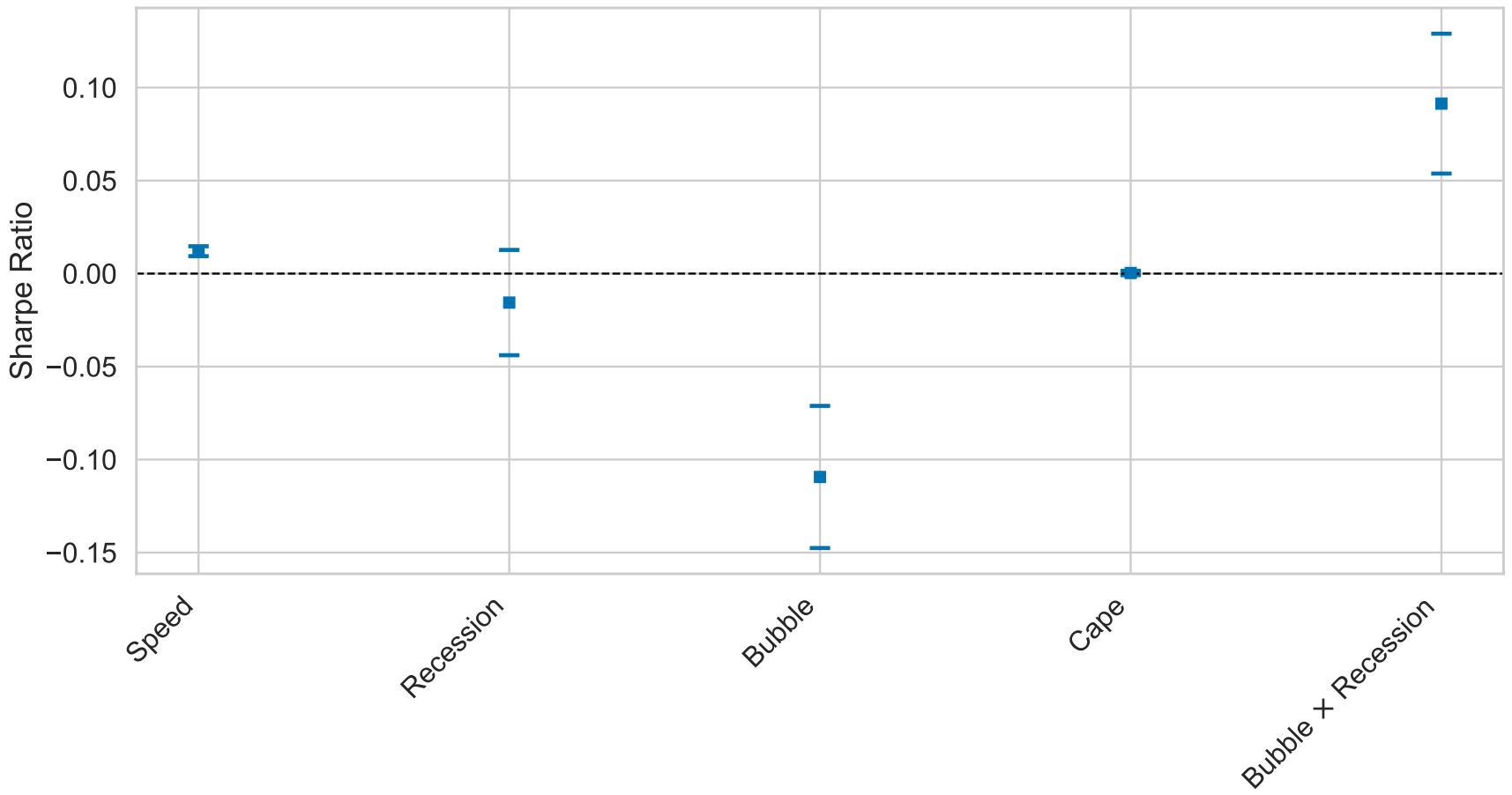


Figure 7: Marginal Effect on the Sharpe Ratio of the Covariates on the Duration (MLE)

Notes: Sensitivity of the expected Sharpe Ratio on each of the variables, estimated by maximum likelihood. The model contains the following covariates: the *speed* of the crash, measured as the number of days the market takes to lose 10% of its peak value; the level of *cape* from Shiller (2015); the identifier of a *recession*, measured as a dummy variable that takes the value of 1 if the respective peak is within a recession period using the NBER business cycle data; and finally the *bubble* identifier, taking the value of 1 if the respective peak is within a bubble period using the Phillips et al. (2015a) methodology; *bubble_and_recession* represents the interaction of both dummies. The sensitivity is calculated by partial derivatives of the Sharpe Ratio with respect to each of the covariates, conditional on the mean value of all other variables. The x-axis shows the value of the covariate, while the y-axis shows the effect on the Sharpe Ratio for a unit increase in that variable. The error bars represent the 95% confidence interval of the marginal effect calculated by drawing 1000 samples from the asymptotic distribution of the parameters using the estimated variance-covariance matrix.

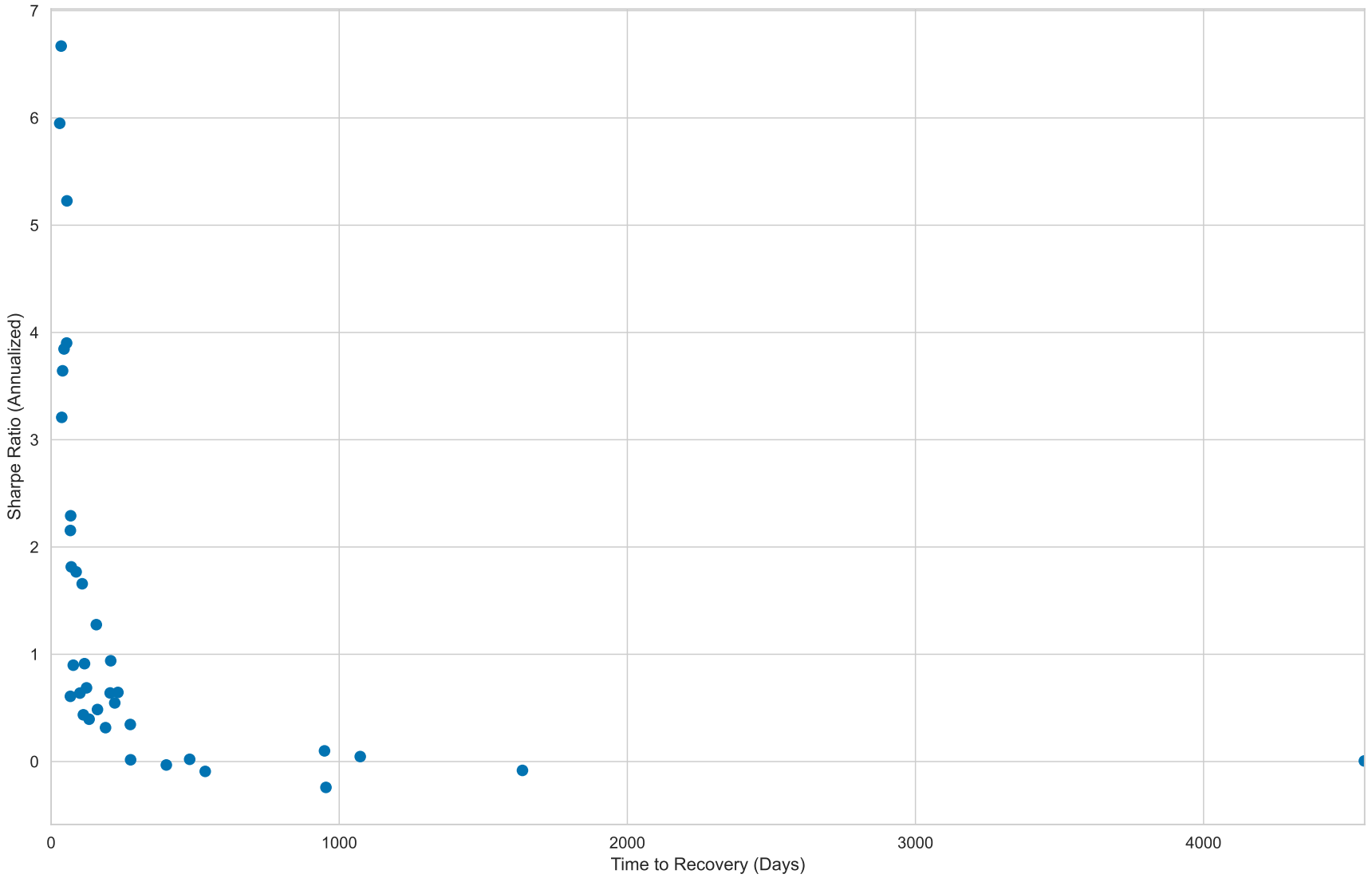


Figure 8: Sharpe Ratio vs. Duration

Notes: Scatter plot of the Sharpe Ratio of the CRSP index during the crash events against the duration of each crash. The Sharpe Ratio is calculated by the average log-return divided by the standard deviation. The final value is then multiplied by $\sqrt{252}$ to annualize. The duration is calculated as the number of days between the drawdown of the crash (the date in which we identify a 10% loss from the peak in the last 252 business days) and the recovery (the date in which the index returns to its pre-crash peak level).

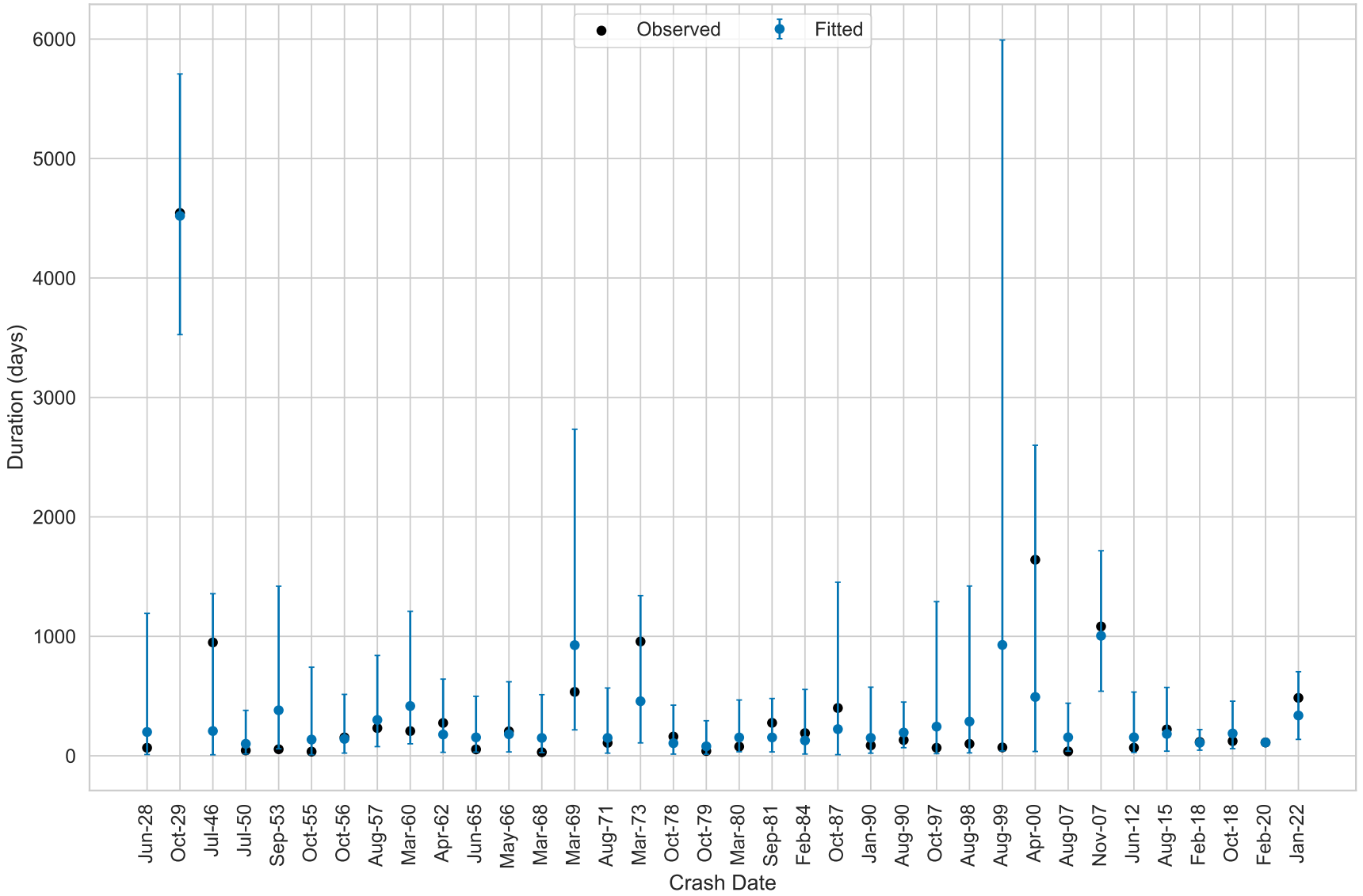


Figure 9: Fitted Values for the Duration of Crashes (MLE)

Notes: Fitted values for the duration of each crash in the CRSP index according to the maximum likelihood estimation (MLE) of the model as shown in Table 5. The observed durations are represented by the black dots and the fitted values are represented by the blue line. The blue shaded area represents the 95% confidence interval of the fitted values. The x-axis shows the crash detection month for each observation in Table 2. The y-axis shows the duration of each crash in days.

B Tables

Table 1: Crashes in Monthly Data

	Drawdown	Bottom	Recovery	Duration	Time to Bottom	Loss to Bottom	Duration P-Value
1	1929-10-01	1932-06-01	1944-10-01	180	32	82.85%	0.50%
2	1929-11-01	1929-11-01	1930-03-01	4	0	12.36%	75.60%
3	1930-06-01	1932-06-01	1936-11-01	77	24	78.82%	2.90%
4	1930-09-01	1932-06-01	1936-07-01	70	21	75.86%	3.50%
5	1931-05-01	1932-06-01	1935-08-01	51	13	66.23%	6.00%
6	1931-09-01	1932-06-01	1935-07-01	46	9	63.65%	6.80%
7	1931-12-01	1932-06-01	1933-05-01	17	6	47.92%	26.80%
8	1932-03-01	1932-06-01	1932-08-01	5	3	42.33%	68.40%
9	1932-04-01	1932-06-01	1932-08-01	4	2	35.12%	75.60%
10	1932-05-01	1932-06-01	1932-07-01	2	1	20.97%	91.20%
11	1932-10-01	1933-02-01	1933-04-01	6	4	26.98%	62.30%
12	1933-02-01	1933-02-01	1933-04-01	2	0	15.41%	91.20%
13	1933-09-01	1933-10-01	1934-01-01	4	1	18.07%	75.60%
14	1934-07-01	1934-07-01	1934-11-01	4	0	10.96%	75.60%
15	1937-09-01	1938-03-01	1943-03-01	66	6	44.32%	3.90%
16	1938-03-01	1938-03-01	1938-06-01	3	0	23.86%	83.60%
17	1939-03-01	1939-04-01	1939-09-01	6	1	12.25%	62.30%
18	1940-05-01	1942-04-01	1943-01-01	32	23	27.98%	12.60%
19	1946-09-01	1947-05-01	1948-05-01	20	8	13.59%	22.20%
20	1970-04-01	1970-06-01	1970-12-01	8	2	20.61%	50.40%

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Table 1: Crashes in Monthly Data

	Drawdown	Bottom	Recovery	Duration	Time to Bottom	Loss to Bottom	Duration P-Value
21	1973-11-01	1974-09-01	1976-02-01	27	10	41.08%	16.10%
22	1974-09-01	1974-09-01	1974-10-01	1	0	11.00%	98.10%
23	1978-10-01	1978-10-01	1979-03-01	5	0	11.23%	68.40%
24	1980-03-01	1980-03-01	1980-06-01	3	0	11.72%	83.60%
25	1987-10-01	1987-11-01	1989-04-01	18	1	28.41%	25.10%
26	1998-08-01	1998-08-01	1998-11-01	3	0	15.71%	83.60%
27	2000-11-01	2002-09-01	2006-01-01	62	22	40.87%	4.90%
28	2002-09-01	2002-09-01	2002-11-01	2	0	10.22%	91.20%
29	2008-10-01	2009-02-01	2010-03-01	17	4	35.76%	26.80%
30	2009-02-01	2009-02-01	2009-04-01	2	0	10.09%	91.20%
31	2020-03-01	2020-03-01	2020-05-01	2	0	13.26%	91.20%

Notes: Crashes in the CRSP index using monthly data. A crash is detected when the market falls by more than 10% from its previous month. The table shows the start date of each crash, the duration (in months), the time from drawdown to bottom (in months) and the total loss to bottom (in %). The last column shows the p-value of the duration, calculated as the proportion of simulated durations that are greater than the observed duration. The p-values are calculated using the empirical distribution of returns, simulated by drawing 1000 *i.i.d.* samples with replacement from the empirical distribution of returns and calculating the time to recover from a 10% loss from the peak (set as 0).

Table 2: Summary of CRSP Index Crashes

	Peak	Drawdown	Bottom	Recovery	Duration (days)	Time to Bottom	Loss to Bottom	Duration P-Value
1	1928-05-14	1928-06-12	1928-06-12	1928-08-30	67	0	10.84%	69.7%
2	1929-09-03	1929-10-04	1932-07-08	1944-12-16	4544	1008	182.64%	0.2%
3	1946-05-29	1946-07-23	1947-05-17	1949-12-05	949	298	33.1%	7.4%
4	1950-06-12	1950-07-11	1950-07-13	1950-09-13	45	2	13.56%	80.3%
5	1953-03-19	1953-09-14	1953-09-14	1953-12-04	55	0	11.49%	75.3%
6	1955-09-23	1955-10-11	1955-10-11	1955-12-01	35	0	10.13%	86.0%
7	1956-08-02	1956-10-01	1957-02-12	1957-05-14	154	134	12.19%	44.5%
8	1957-07-15	1957-08-26	1957-10-22	1958-07-28	232	57	23.03%	32.4%
9	1960-01-05	1960-03-07	1960-10-25	1960-12-30	207	232	11.03%	35.4%
10	1961-12-12	1962-04-30	1962-06-26	1963-06-03	275	57	32.42%	28.5%
11	1965-05-13	1965-06-28	1965-06-28	1965-09-14	54	0	10.13%	75.7%
12	1966-02-09	1966-05-17	1966-10-07	1967-03-10	205	143	23.04%	35.5%
13	1968-01-12	1968-03-05	1968-03-05	1968-04-18	30	0	10.35%	90.6%
14	1968-11-29	1969-03-14	1970-05-26	1971-04-28	535	438	45.92%	14.8%
15	1971-04-28	1971-08-04	1971-11-23	1972-01-06	108	111	13.9%	55.7%
16	1973-01-11	1973-03-21	1974-10-03	1977-01-03	957	561	66.18%	7.3%
17	1978-09-12	1978-10-23	1978-11-14	1979-06-13	161	22	16.43%	43.2%
18	1979-10-05	1979-10-22	1979-10-25	1979-12-18	40	3	11.51%	83.8%
19	1980-02-13	1980-03-07	1980-03-27	1980-06-26	77	20	20.7%	66.0%
20	1981-08-11	1981-09-04	1982-08-12	1982-10-08	276	342	22.47%	28.5%
21	1983-06-22	1984-02-08	1984-07-24	1984-11-07	190	167	14.69%	38.3%

Continued on next page

Table 2: Summary of CRSP Index Crashes

	Peak	Drawdown	Bottom	Recovery	Duration (days)	Time to Bottom	Loss to Bottom	Duration P-Value
22	1987-08-25	1987-10-15	1987-12-04	1989-05-16	400	50	40.39%	20.4%
23	1989-10-09	1990-01-25	1990-01-30	1990-05-31	87	5	11.65%	62.0%
24	1990-07-16	1990-08-06	1990-10-11	1991-02-12	132	66	23.32%	49.7%
25	1997-10-07	1997-10-27	1997-10-27	1998-02-03	67	0	10.47%	69.7%
26	1998-07-17	1998-08-04	1998-10-08	1998-12-24	100	65	24.77%	57.7%
27	1999-07-16	1999-08-10	1999-10-15	1999-11-17	70	66	12.12%	68.0%
28	2000-03-24	2000-04-13	2002-10-09	2006-10-24	1641	909	68.56%	3.5%
29	2007-07-13	2007-08-15	2007-08-15	2007-10-08	37	0	10.13%	85.2%
30	2007-10-09	2007-11-26	2009-03-09	2012-03-14	1083	469	79.15%	6.1%
31	2012-04-02	2012-06-01	2012-06-04	2012-09-07	68	3	10.74%	68.9%
32	2015-06-23	2015-08-24	2016-02-11	2016-07-11	221	171	17.03%	33.6%
33	2018-01-26	2018-02-08	2018-02-08	2018-07-26	116	0	10.41%	53.4%
34	2018-09-20	2018-10-24	2018-12-24	2019-04-24	123	61	22.98%	52.2%
35	2020-02-19	2020-02-27	2020-03-23	2020-08-06	112	25	41.97%	54.6%
36	2021-11-16	2022-01-21	2022-10-14	2023-12-27	485	266	29.97%	17.1%

Notes: This table shows the crash events identified in the CRSP index since 1926. The crashes are defined as periods in which the index falls by more than 10% from its last peak (maximum index value in the last 252 business days), ending when the index returns to its pre-crash peak level. The table shows the date of the peak, the drawdown (when the 10% fall is first detected), the date of the bottom (where the index reaches its minimum for the time span of the crash), the date of the recovery, the duration of the crash (days between the recovery and the drawdown) as well as the time to bottom (days between the Drawdown and the Bottom columns) and the total loss to bottom (in %). The last column shows the p-value of the duration, calculated as the proportion of simulated durations that are greater than the observed duration. The p-values are calculated using the empirical distribution of returns, simulated by drawing 1000 *i.i.d.* samples with replacement from the empirical distribution of returns and calculating the time to recover from a 10% loss from the peak (set as 0).

Table 3: Kolmogorov-Smirnov test for Simulated and Empirical Duration against Inverse Gaussian

		Critical Value		
	Test Statistic	10%	5%	1%
Empirical	0.9987	0.3752	0.3959	0.4358
Simulated	1.0000	0.2485	0.2569	0.2727

Notes: Results of the Kolmogorov-Smirnov test to compare the empirical and simulated distribution of duration to the theoretical (Inverse Gaussian) distribution. The empirical distribution is calculated from the observed durations in the CRSP index. The simulated distribution is calculated by drawing *i.i.d.* samples from the empirical distribution of returns and calculating the time to reach a 10% gain from the peak (set as 0). For the Inverse Gaussian distribution, the moments are fitted using the Method of Moments to the empirical mean and variance of the returns. Due to the estimation of the parameters, the critical values need to be adjusted. We use the results of Gunes et al. (1997) to calculate these critical values.

Table 4: Duration Regression Results

	Dependent variable: duration				
	(1)	(2)	(3)	(4)	(5)
intercept	-157.628	-628.996	140.917	-314.345	12.779
	(414.973)	(415.511)	(439.830)	(410.123)	(204.175)
speed	26.413	65.976	-95.907	-79.847	-63.538
	(155.655)	(142.615)	(167.500)	(146.755)	(72.090)
speed^2	-6.471	-12.680	3.800	-0.754	1.389
	(13.398)	(12.421)	(14.353)	(12.647)	(6.215)
cape	24.802*	34.134**	14.594	22.629	12.391*
	(14.501)	(13.650)	(15.318)	(13.642)	(6.781)
recession		759.275**		842.854***	315.852**
		(276.994)		(260.980)	(139.061)
bubble			616.469*	756.868**	272.835
			(362.673)	(320.535)	(165.030)
bubble × recession					3743.539***
					(383.290)
Observations	36	36	36	36	36
R^2	0.095	0.271	0.172	0.385	0.857
Adjusted R^2	0.010	0.177	0.065	0.283	0.827
Residual Std. Error	793.446 (df=32)	723.243 (df=31)	771.012 (df=31)	675.133 (df=30)	331.554 (df=29)
F Statistic	1.114 (df=3; 32)	2.884** (df=4; 31)	1.608 (df=4; 31)	3.763*** (df=5; 30)	28.902*** (df=6; 29)

Notes: Regression results for the reduced form duration model. The dependent variable is the duration of the observed crashes in the CRSP index. The covariates are: the *speed* of the crash, measured as the number of days the market takes to lose 10% of its peak value; the level of *cape* from Shiller (2015); the identifier of a *recession*, measured as a dummy variable that takes the value of 1 if the respective peak is within a recession period using the NBER business cycle data; and finally the *bubble* identifier, taking the value of 1 if the respective peak is within a bubble period using the Phillips et al. (2015a) methodology; *bubble_and_recession* represents the interaction of both dummies. For each of the covariates, the coefficient is shown, together with the robust standard deviation in parenthesis and the significance level in stars ($^*p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$).

Table 5: Maximum Likelihood Estimation Results

	Mean	Vol
	(1)	(2)
Intercept	23.7314***	21.1136***
	(2.5079)	(2.9226)
Speed	2.0964***	0.0179
	(0.0029)	(0.1396)
Recession	-11.8260***	-7.3508**
	(2.4892)	(3.2806)
Bubble	-7.3206	9.2538**
	(4.5299)	(3.9750)
Cape	-0.5038***	-0.4497***
	(0.0266)	(0.0515)
Bubble × Recession	5.6163	-8.1384**
	(4.5380)	(4.0233)
Observations	36	36
Log-likelihood	-6.0276	-6.0276

*p<0.1; **p<0.05; ***p<0.01

Notes: Maximum Likelihood Estimation Results for the duration model. The table shows the coefficients and the standard errors below in parenthesis. The covariates are: the *speed* of the crash, measured as the number of days the market takes to lose 10% of its peak value; the level of *cape* from Shiller (2015); the identifier of a *recession*, measured as a dummy variable that takes the value of 1 if the respective peak is within a recession period using the NBER business cycle data; and finally the *bubble* identifier, taking the value of 1 if the respective peak is within a bubble period using the Phillips et al. (2015a) methodology; *bubble_and_recession* represents the interaction of both dummies. The first column displays the coefficients for the mean returns and the second column shows the coefficients of the return volatility for each of the covariates. All coefficients and standard errors are in percentage terms, annualized by multiplying by 252 for the mean returns and by $\sqrt{252}$ for the volatility.

Table 6: Maximum Likelihood Estimation Strategy Results

		Short Sell			No Short Sell		
	Market	Bubble	Recession	Speed	Bubble	Recession	Speed
Average	9.33%	7.83%	7.29%	6.06%	8.58%	8.31%	6.88%
Std. Dev.	17.15%	17.15%	17.15%	11.17%	12.00%	11.13%	9.88%
Q25%	-98.23%	-103.36%	-104.81%	-27.74%	-31.27%	-27.23%	0.00%
Median	18.90%	12.09%	12.60%	0.00%	0.00%	0.00%	0.00%
Q75%	128.45%	125.21%	124.45%	62.94%	72.48%	68.45%	38.28%
Alpha		4.87%***	5.19%***	1.47%	2.44%***	2.59%***	1.72%**
		(2.880)	(3.104)	(1.439)	(2.880)	(3.104)	(2.164)
Beta		-0.020***	-0.157***	0.240***	0.490***	0.421***	0.332***
		(-3.271)	(-25.606)	(63.712)	(157.677)	(137.331)	(113.332)
Signal		−	−	+	−	−	+
Obs	25901	25901	25901	25901	25901	25901	25901
Sharpe Ratio	0.364	0.277	0.245	0.266	0.458	0.469	0.384
Obs (Exposed)	25901	13974	13974	5235	6092	5753	2329
Sharpe Ratio (Exposed)	0.364	0.022	-0.026	-0.277	0.318	0.370	0.217

*p<0.1; **p<0.05; ***p<0.01

Notes: Descriptive statistics for the strategies using the maximum likelihood results. The table shows the average return (annualized by multiplying by 252), standard deviation and Sharpe ratio (annualized by multiplying by $\sqrt{252}$) of each strategy both for the full sample (which includes the periods in which each strategy is not invested) and for the exposed sample (denoted by the Non-Zero Exposure columns) where the strategy is indeed invested in the market.

Table 7: Cumby-Modest Test for Market Timing Strategies

	Dependent variable: Excess Return					
	Speed	Bubble	Recession	Speed (no SS)	Bubble (no SS)	Recession (no SS)
α	2.386	3.036	2.317	2.756	1.384	2.120
	(2.685)	(2.690)	(2.651)	(2.773)	(2.798)	(2.897)
β	-1.844	4.464*	1.797	-4.386	3.758	0.257
	(4.313)	(2.558)	(2.480)	(6.427)	(4.242)	(4.311)
Observations	25650	25650	25650	25650	25650	25650
R^2	0.002	0.002	0.002	0.002	0.002	0.002
Adjusted R^2	0.001	0.001	0.001	0.001	0.001	0.001
Residual Std. Error	272.785	272.770	272.783	272.784	272.782	272.786
F Statistic	2.331***	2.461***	2.346***	2.344***	2.358***	2.322***
Note:						*p<0.1; **p<0.05; ***p<0.01

Notes: Results for the Cumby-Modest test for market timing strategies. The test involves regressing the excess return of the market on a cosntant, the strategy weights (as binary) and additional factors for each of the strategies. The factors included are the autocorrelation of the market excess returns up to 21 lags (hidden for simplicity), using a window size of 252. Numbers in parenthesis show the standard errors for the coefficients estimated via HAC. The significance of the strategy is equivalent to the significance of the strategy weights. Returns are shown in percentage terms, annualized by multiplying by 252.

Table 8: Summary Statistics of Crash Strategies

		Full Sample			Non-Zero Exposure			
		Average	Std. Dev.	Sharpe Ratio	Count	Average	Std. Dev.	Sharpe Ratio
Pro	All Crashes	1.75	15.37	0.11	13905	3.26	20.98	0.16
	No Outliers	1.75	8.68	0.20	6661	6.82	17.12	0.40
Anti		4.50	7.62	0.59	11840	9.84	11.26	0.87

Notes: Performance of the 3 strategies developed in Section 5. The pro-crash strategy buys the market excess returns (long in the market and short in the risk-free rate) after a crash is identified and holds it until recovery, when it divests. The anti-crash strategy does the opposite, buying the excess returns when the crash ends and selling them once another crash begins. Finally, the modified pro-crash strategy in the second line takes a *naive* approach to remove the 3 most extreme crashes (in terms of duration) from the account, not investing in the market excess returns during these periods. The table then shows the average return (annualized by multiplying by 252), standard deviation and Sharpe ratio (annualized by multiplying by $\sqrt{252}$) of each strategy both for the full sample (which includes the periods in which each strategy is not invested) and for the invested sample (denoted by the Non-Zero Exposure columns) where the strategy is indeed invested in the market. All statistics are considered in log-returns.

Table 9: Crash Strategy Regression

	Pro				Anti	
	All Crashes		No Outliers			
	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	-6.004***	-5.068***	-0.297	-0.074	2.921***	1.985***
	(0.665)	(0.646)	(0.724)	(0.726)	(0.665)	(0.646)
MKT	0.793***	0.783***	0.271***	0.269***	0.207***	0.217***
	(0.002)	(0.002)	(0.003)	(0.003)	(0.002)	(0.002)
SMB	-0.005	-0.007	0.020***	0.019***	0.005	0.007*
	(0.004)	(0.004)	(0.005)	(0.005)	(0.004)	(0.004)
HML	0.114***	0.085***	-0.155***	-0.162***	-0.114***	-0.085***
	(0.004)	(0.004)	(0.005)	(0.005)	(0.004)	(0.004)
MOM		-0.134***		-0.031***		0.133***
		(0.003)		(0.004)		(0.003)
Observations	25901	25800	25901	25800	25901	25800
R^2	0.808	0.820	0.288	0.290	0.219	0.264
Adjusted R^2	0.808	0.820	0.287	0.290	0.219	0.264
Residual Std. Error	106.905	103.628	116.373	116.412	106.893	103.633
F Statistic	36331.552***	29440.337***	3483.492***	2632.053***	2416.192***	2311.708***

*p<0.1; **p<0.05; ***p<0.01

Notes: Regression results for the performance of each of the 3 strategies developed in Section 5 comparing with the Fama-French 3 factors + Momentum. The pro-crash strategy buys the market excess returns (long in the market and short in the risk-free rate) after a crash is identified and holds it until recovery, when it divests. The anti-crash strategy does the opposite, buying the excess returns when the crash ends and selling them once another crash begins. Finally, the modified pro-crash strategy in the second line takes a *naive* approach to remove the 3 most extreme crashes (in terms of duration) from the account, not investing in the market excess returns during these periods. The table shows the regression result for each of these strategies against the market excess-returns (MKT), the size factor (SMB), the value factor (HML). The right column for each strategy also includes the additional momentum factor (MOM). For each of the covariates, the coefficient is shown, together with the standard deviation in parenthesis and the significance level in stars.

C Model

In this section we verify the necessary and sufficient conditions for the ML estimator to be consistent and asymptotically normal.

Recall the log-likelihood function given by:

$$\ell(\beta,\gamma) \propto -\log\left(\gamma' F_i\right) - \frac{\left(\alpha - \left(\beta' F_i\right) T_i\right)^2}{2\left(\gamma' F_i\right)^2 T_i} \tag{18}$$

The gradient of ℓ w.r.t the parameters is

$$\nabla\ell(\beta,\gamma) = \begin{pmatrix} \frac{\partial\ell}{\partial\beta} \\ \frac{\partial\ell}{\partial\gamma} \end{pmatrix} = \begin{bmatrix} \frac{\alpha-(\beta' F)T}{(\gamma' F)^2 T} \\ -\frac{1}{\gamma' F} + \frac{[\alpha-(\beta' F)T]^2}{(\gamma' F)^3 T} \end{bmatrix} \otimes F \tag{19}$$

and the Hessian is

$$\mathcal{H}(\beta,\gamma) \coloneqq \begin{bmatrix} \frac{\partial^2\ell}{\partial\beta^2} & \frac{\partial\ell}{\partial\beta\gamma} \\ \frac{\partial\ell}{\partial\gamma\beta} & \frac{\partial^2\ell}{\partial\gamma^2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{(\gamma' F)^2} & -\frac{2(\alpha-(\beta' F)T)}{(\gamma' F)^3 T} \\ -\frac{2(\alpha-(\beta' F)T)}{(\gamma' F)^3 T} & \frac{1}{(\gamma' F)^2} - \frac{3(\alpha-(\beta' F)T)^2}{(\gamma' F)^4 T} \end{bmatrix} \otimes (FF') \tag{20}$$

Assuming that FF' is positive definite, the conditions for $\mathcal{H}(\beta,\gamma)$ to be negative definite are:

$$\frac{1}{\left(\gamma' F\right)^2} > 0 \iff \gamma' F \neq 0 \tag{21}$$

$$\frac{1}{\left(\gamma' F\right)^2} - \frac{3\left(\alpha - \left(\beta' F\right) T\right)^2}{\left(\gamma' F\right)^4 T} < 0 \iff \left(\gamma' F\right)^2 T \leq 3\left(\alpha - \left(\beta' F\right) T\right)^2 \tag{22}$$

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